

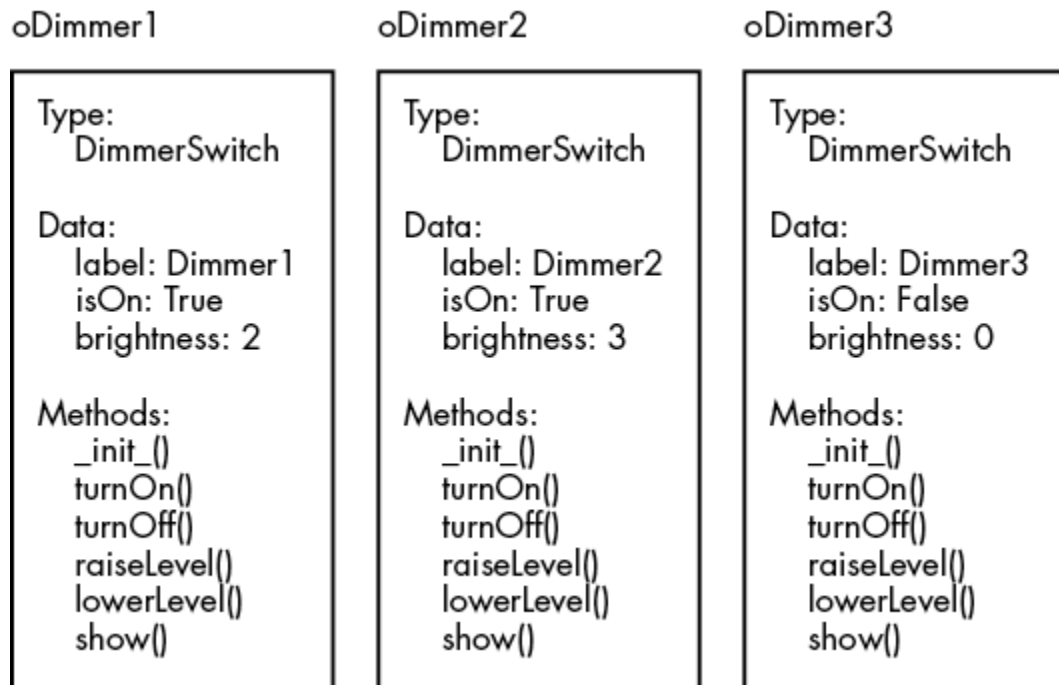


Figure 3-1: In mental model #1 each object is a unit that has a type, data, and methods.

The data and methods of each object are packaged together. The scope of an instance variable is defined as all the methods in the class, so all methods have access to the instance variables associated with that object.

If this mental model makes the concepts clear, then you're in good shape. While this is *not* the way objects are actually implemented, it's a perfectly reasonable way to *think about* how an object's instance variables and methods work together.

A Deeper Mental Model #2

This second model explores objects at a lower level and will explain more about what an object is.

Every time you instantiate an object, you get back a value from Python. We typically store the returned value in a variable that refers to the object. In [Listing 3-2](#), we create three `DimmerSwitch` objects. After creating each one, we'll add code to inspect the result by printing out the type and value of each variable.